

Case Study: Strategic Application of Technical Indicators in Financial Markets

A Comprehensive Analysis of Moving Averages, RSI, and MACD

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1. Introduction

Technical analysis is a trading discipline employed to evaluate investments and identify trading opportunities by analyzing statistical trends gathered from trading activity, such as price movement and volume. Unlike fundamental analysis, which assesses a company's intrinsic value through earnings and macro-economic health, technical analysis focuses on market psychology and historical price patterns. This case study explores how three foundational tools—Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Moving Averages (MA)—can be integrated to form a disciplined trading framework.

2. Moving Averages: The Trend Filter

Moving averages smooth out price fluctuations to reveal the underlying direction of a market.

- **Significance:** They act as dynamic support or resistance zones.
- **Strategy:** Traders often use a 200-day Simple Moving Average (SMA) as a "trend filter." A price above the 200-SMA suggests a long-term uptrend, while a price below suggests a downtrend.
- **Crossover Signals:** When a short-term MA (e.g., 50-day) crosses above a long-term MA (e.g., 200-day), it is known as a "Golden Cross," signaling strong bullish momentum.



3. RSI: Gauging Momentum

The RSI is a momentum oscillator that measures the speed and change of price movements on a scale of 0 to 100.

- **Overbought/Oversold:** Traditionally, $RSI > 70$ indicates an overbought condition (potential for pullback), and $RSI < 30$ indicates an oversold condition (potential for bounce).
- **Divergence:** A "Bullish Divergence" occurs when a stock hits a new low but the RSI makes a higher low, suggesting that downward momentum is fading and a reversal may be near.



4. MACD: Confirming Momentum Shifts

The MACD helps traders identify the strength and direction of a trend.

- **Components:** It consists of the MACD line (difference between 12-day and 26-day EMA), a signal line (9-day EMA), and a histogram.
- **Crossovers:** A bullish signal is generated when the MACD line crosses above the signal line.
- **Zero-Line Cross:** When the MACD line crosses above the zero line, it indicates that short-term momentum has overtaken long-term momentum, confirming a trend change.



5. Integrated Strategy: The "Dynamic Trio"

Success in technical analysis relies on the confluence of multiple indicators.

- **The Framework:**
 1. **Identify Trend:** Use the 200-day MA to confirm the long-term trend direction.
 2. **Confirm Momentum:** Wait for the MACD line to cross the signal line in the direction of the trend.
 3. **Time the Entry:** Use the RSI to ensure the entry is not made at an "overbought" level.
- *Note to Student:* **[Insert Chart Here]** – Take a screenshot from TradingView or Moneybhai showing a stock price chart with these three indicators applied. Annotate the points where all three indicators aligned to create a buy signal.



6. Conclusion and References

While technical indicators are powerful, they are not foolproof and are prone to "noise" in volatile markets. They should be used to provide structure rather than as a guaranteed prediction of future price movement.

References

- Grip Invest. "Technical Indicators Explained: RSI, MACD & Moving Average Strategy."
- Doon Trading Academy. "Technical Indicators RSI, MACD and Moving Averages – Complete Usage Guide."
- Study.com. "Common Technical Analysis Indicators."
- Phillip Nova. "RSI, MACD, EMA: High-Probability Setups for Forex Traders." .